

Joshua Sternadel

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Systems Architecture & Deep-Infrastructure Engineering

A high-capacity Systems Architect who leverages advanced spatial thinking and deep pattern recognition to solve complex structural debt that traditional engineering groups cannot resolve. A self-directed technical anchor specializing in building clean, frictionless distributed systems from first principles. Proven track record of delivering radical automation, continuous DevSecOps compliance, and mission-critical stability across high-scale enterprise environments.

Core Competencies

Team Leadership | Systems Architecture | Solutions Engineering | Efficiency and Productivity Specialist

Key Achievements

- **Strategic Retention & Continuity:** Awarded a significant retention bonus during the Sprint/T-Mobile merger, designated as mission-critical personnel to ensure technical stability and architectural integration during the corporate transition.
- **U.S. Patent Author (#10,366,066):** Co-authored a patent for database management systems, specifically engineering a natural sorting algorithm in SQL to optimize data retrieval and user experience.
- **Infrastructure Transformation:** Orchestrated a full-scale migration from legacy physical servers to a redundant VMware-based virtual infrastructure, reducing enterprise downtime by over 90% and cutting operational payroll costs by 18%.
- **Algorithmic Solutions:** Engineered a first-principles solution to resolve GPS signal noise and "teleportation" errors for intermodal logistics. By deciphering raw NMEA 0183 sentences and applying spherical geometry with rate-of-acceleration modeling, successfully stabilized coordinate paths and eliminated tracking jitter within high-interference rail yards.

Professional Experience

T-Mobile | Senior Software Engineer | Acting Solutions Architect | 2021 to 2026

Embedded within Tools Team to assist with architectural duties and build APIs that connect proprietary systems to Atlassian and other vendors' products. Transitioned to DevOps role as the only true developer on the SysAdmin team.

- **Enterprise Cloud Migration:** Architected and executed a high-integrity migration of a decade's worth of metadata and complex workflows from Jira Data Center to Jira Cloud, ensuring zero data loss across mission-critical proprietary projects.

- **Ecosystem Integration & APIs:** Authored a specialized suite of Node.js libraries and Node-RED nodes to automate full-lifecycle project provisioning via Atlassian APIs, creating a “self-service” ecosystem for developers.
- **Infrastructure as Code (IaC):** Bridged the gap between Software Engineering and Systems Administration by implementing Infrastructure as Code (IaC) via Ansible, successfully reducing manual operational toil and human error.
- **DevSecOps & Automation:** Engineered automated vulnerability remediation workflows that drove lab data center security debt from 100+ monthly criticals to zero, achieving a state of continuous compliance.
- **Value Stream Optimization:** Optimized the VM lifecycle by automating the end-to-end onboarding process, significantly accelerating the “Lead Time to Deployment” for internal engineering teams.

Sprint | Application Developer III | Acting Principal Engineer | 2017 to 2021

Served as Core Technical Advisor and Architectural Gatekeeper to leadership, maintaining absolute technical veto power and engineering oversight for all departmental systems and cross-functional integrations.

- **Architectural Leadership:** Directed the development of a “First Principles” replacement for legacy SKU management, delivering a high-performance solution in 6 months that outpaced a decade of failed initiatives by multiple cross-functional teams.
- **Distributed Systems:** Designed “Porcupine,” a custom message bus system enabling real-time, asynchronous communication between disparate enterprise applications.
- **Framework Engineering:** Engineered a custom, data-agnostic PHP framework utilizing reflection and annotations to automate the CRUD lifecycle across disparate schemas (SQL, NoSQL, JSON, XML).
- **Data Abstraction & Hydration:** Developed a model-driven hydration engine that abstracted away storage-layer complexity, allowing for seamless state transitions between relational and non-relational data sources.
- **Scale & Adoption:** Prototyped and deployed the framework to internal high-value T-Mobile applications, proving a 10x reduction in “boilerplate” code for new service onboarding.
- **Site Reliability:** Owned and managed a complex 42-VM ecosystem, maintaining high availability and scalability for the Device and Technology Organization.
- **Intellectual Property:** Co-authored a U.S. Patent for a novel SQL data storage method, optimizing natural sorting algorithms for massive datasets.

Sprint | Product Development Engineer II | 2013 to 2017

Transitioned from device testing into application development through hybrid role bridging both disciplines.

- **Full-Stack Application Architecture:** Engineered a specialized visual prototyping platform featuring a drag-and-drop grid system to simulate and design Out-of-Box Experiences (OOBE); integrated background profiling engines to capture real-time CPU, RAM, storage, and battery metrics alongside automated Play Store data scraping.
- **Database Engineering & Tooling:** Authored the foundational codebase for the Device Development and Certification requirements and test-plan database, laying the groundwork for scalable documentation and testing.
- **Cross-Functional Advisory:** Advised the Retail Department in weekly technical reviews, clarifying bugs, known issues, and system behaviors so teams could plan and prioritize work with greater accuracy.
- **Release Readiness & Prioritization:** Planned and prioritized weekly device onboarding for the Slate platform, improving visibility and accelerating readiness across teams.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012 to 2013

Developed, tested, and troubleshot XML scripts for the SLATE platform, which retail stores use for automated device testing.

- Wrote XML scripts for Slate automation testing of mobile devices.
- Worked with OEM partners, troubleshooting issues with the Slate Device Testing platform.

MO-KAN Container Services | System Administrator | 2011 to 2012

- Managed all technology operations across 3 onsite locations, keeping systems stable and responsive for daily business needs.
- Planned and executed a full network modernization, replacing aging G3 servers with a VMware-based virtual infrastructure and scalable iSCSI SAN.
- Cut IT downtime by more than 90% through a shift to virtualized, redundant systems.
- Reduced IT payroll costs by nearly 18% by upgrading infrastructure and implementing automated, failover-ready environments.

Additional Relevant Experience

Trucking Central | Software Consultant (Contract)

Planned, developed, and tested Android-based GPS solutions for the intermodal supply chain industry.

Innovative Solutions | Mobile Solutions Technician

Troubleshoot and fixed Nextel devices for first responders and construction businesses, listened to client needs, and leveraged emerging technologies to find solutions for those needs.

Community Impact & Leadership

Seek and Serve | *Technical Advisory Board Member* | 2026 to Present

- **Technical Guidance:** Provide high-level architectural oversight and logistical strategy to the parent 501(c)(3) organization.

33meals.org (Subsidiary) | *Infrastructure Consultant / Volunteer* | 2026 to Present

- **Organizational Launch:** Spearheaded the launch and incorporation of 33meals.org as a dedicated subsidiary to address food insecurity through scalable, data-driven, distribution models.
- **Operational Infrastructure:** Managed the full technical and administrative setup, including domain registration, digital architecture, and regulatory filings to ensure high-integrity non-profit operations.
- **Strategic Development:** Designed a streamlined logistics model to optimize charitable food delivery, applying enterprise efficiency principles to humanitarian service.

Education

Associate of Arts (AA), Graphic Design | Johnson County Community College

Technical Environment Matrix

- **Architectural Disciplines:** Systems Architecture, Distributed Systems, Microservices, API Gateway Design, Platform Abstractions, SRE, DevSecOps
- **Frontend Web Frameworks:** Angular (Pre-1.0 to v15), Vue.js 2, TypeScript, JavaScript (ES6+), Client-Side Architecture
- **Backend Environments & Languages:** Node.js Runtime, .NET Core (2.x), C#, Native Modules, Zero-Dependency Design
- **Infrastructure & Automation:** Ansible, DevOps, CI/CD, Infrastructure as Code (IaC), Automation, Site Reliability, Red Hat Linux, VMware
- **Data & Ecosystems:** Dremio (Lakehouse Federation), ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Rocket.Chat, PostgreSQL, SQL, MongoDB, Atlassian SDK, Jira API (US Patent #10,366,066)